Histogram-of-an-images

Aim

To obtain a histogram for finding the frequency of pixels in an Image with pixel values ranging from 0 to 255. Also write the code using OpenCV to perform histogram equalization.

Software Required:

Anaconda - Python 3.7

Algorithm:

Step1:

Read the gray and color image using imread()

Step2:

Print the image using imshow().

Step3:

Use calcHist() function to mark the image in graph frequency for gray and color image.

step4:

Use calcHist() function to mark the image in graph frequency for gray and color image.

Step5:

The Histogram of gray scale image and color image is shown.

Program:

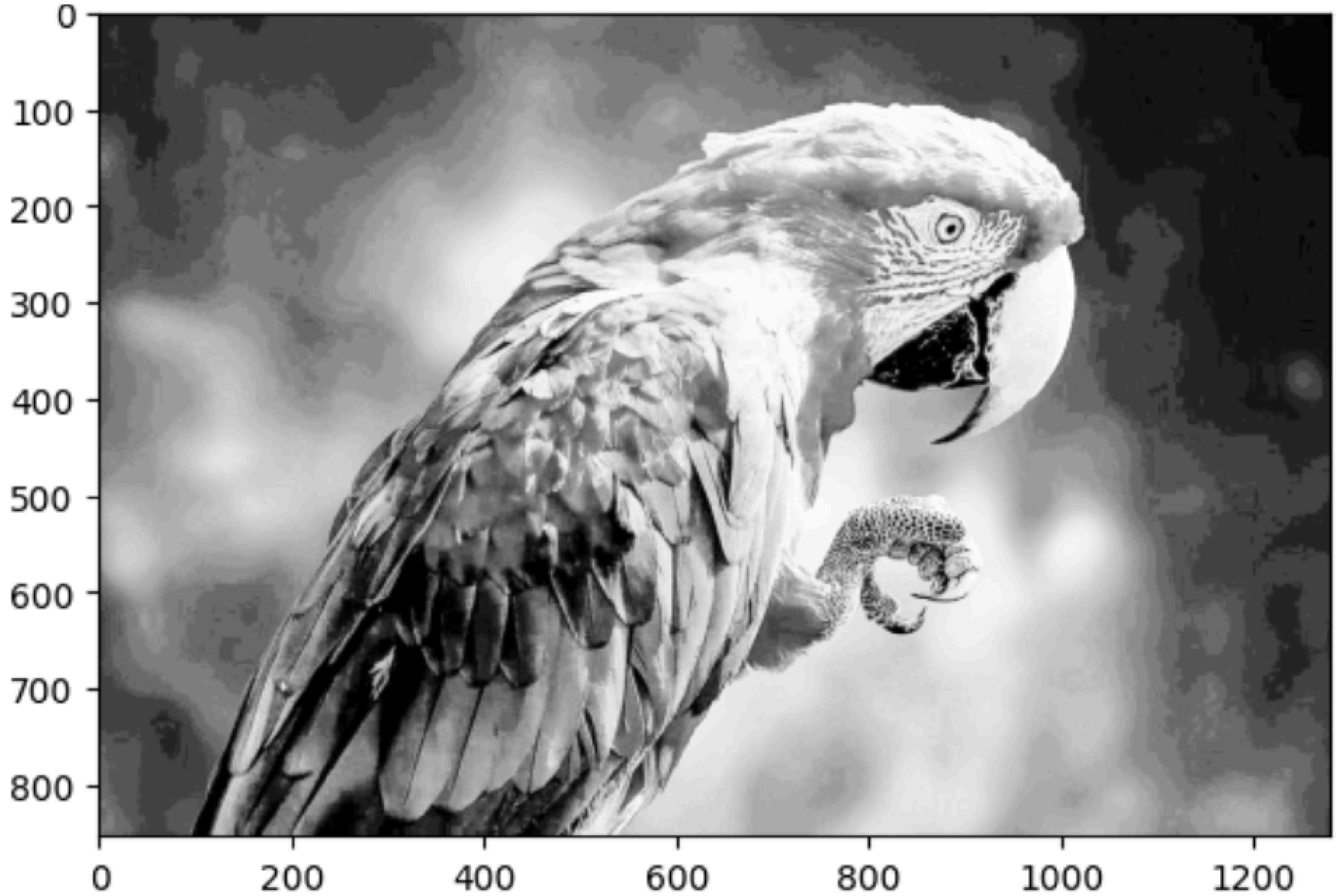
```
# Developed By: HENRIPRASATH S
# Register Number: 212223230077
```

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread('parrot.jpg', cv2.IMREAD_GRAYSCALE)

plt.imshow(img, cmap='gray')
```

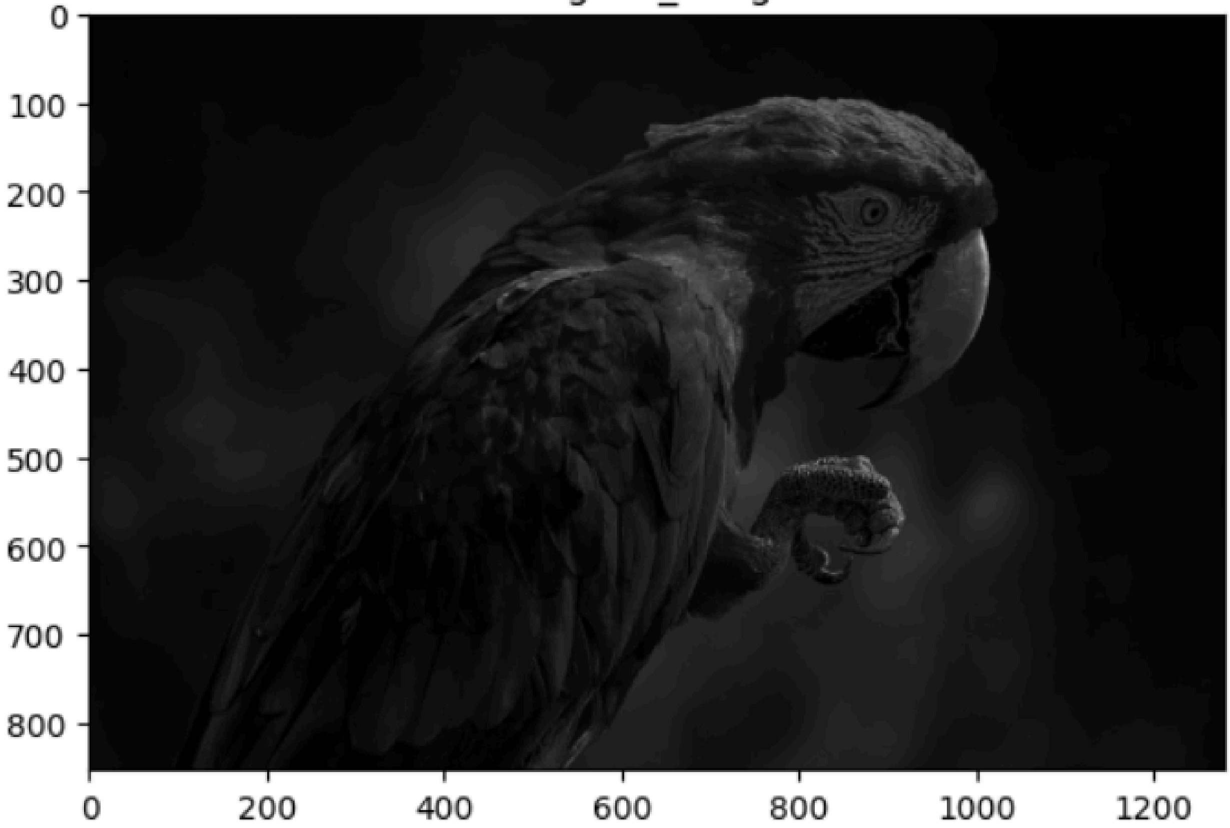
```
plt.title('original_image')
plt.show()

plt.hist(img.ravel(),256,range = [0, 256]);
plt.title('Original Image')
plt.show()

img_eq = cv2.equalizeHist(img)

plt.hist(img_eq.ravel(), 256, range = [0, 256]);
plt.title('Equalized Histogram')

plt.imshow(img_eq, cmap='gray')
plt.title('original image')
plt.show()

img = cv2.imread('parrot.jpg', cv2.IMREAD_COLOR)
img_hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
img_hsv[:,:,2] = cv2.equalizeHist(img_hsv[:, :, 2])
img_eq = cv2.cvtColor(img_hsv, cv2.COLOR_HSV2BGR)

plt.figure(figsize = [12,10])
plt.subplot(221); plt.imshow(img[:, :, ::-1]); plt.title('Original Color Image')
plt.subplot(222); plt.imshow(img_eq[:, :, ::-1]); plt.title('Equalized Image')
plt.subplot(223); plt.hist(img.ravel(),256,range = [0, 256]); plt.title('Original Image')
plt.subplot(224); plt.hist(img_eq.ravel(),256,range = [0, 256]); plt.title('Histogram Equalized'
```

Output:

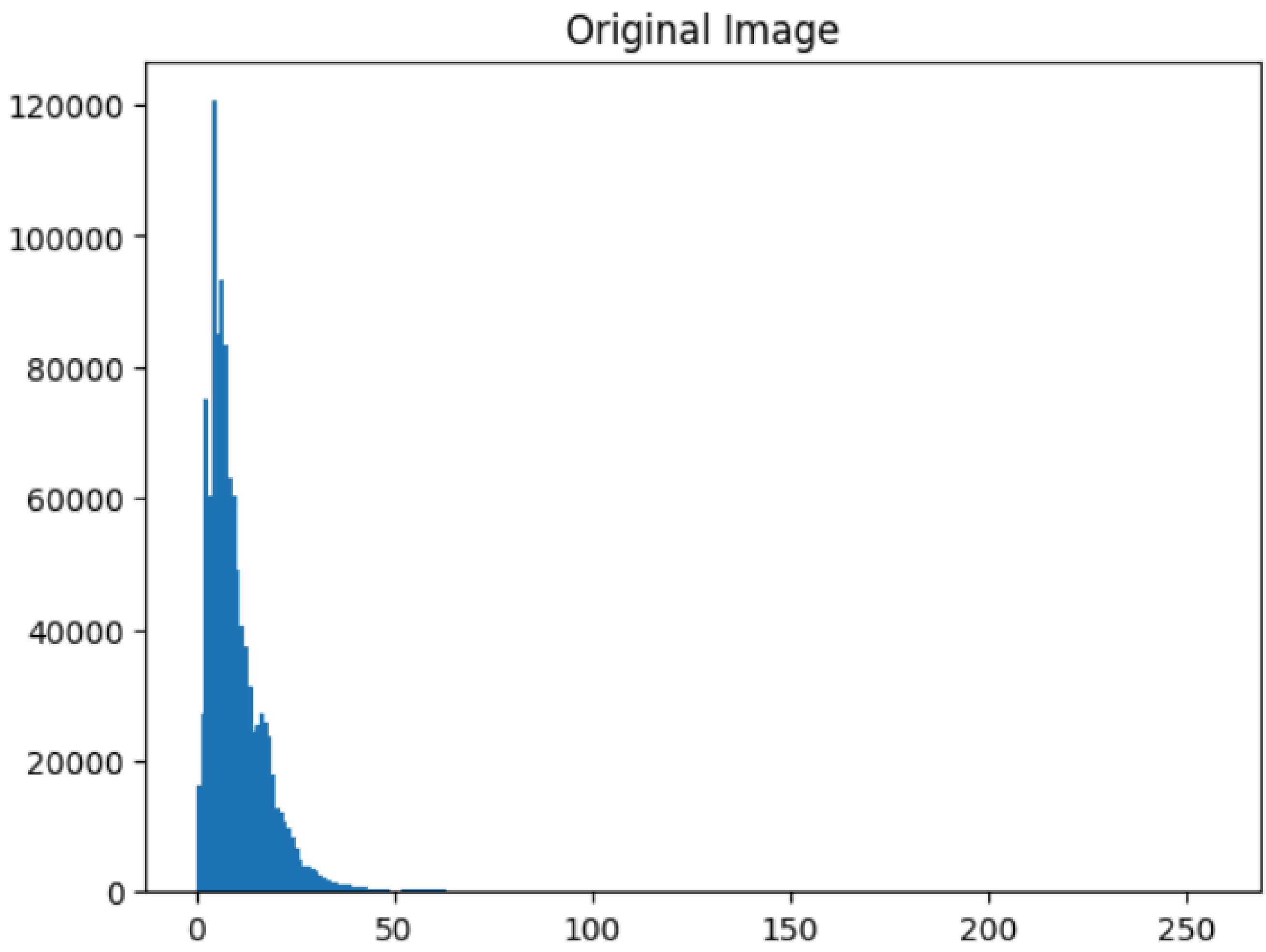
Input Grayscale Image and Color Image



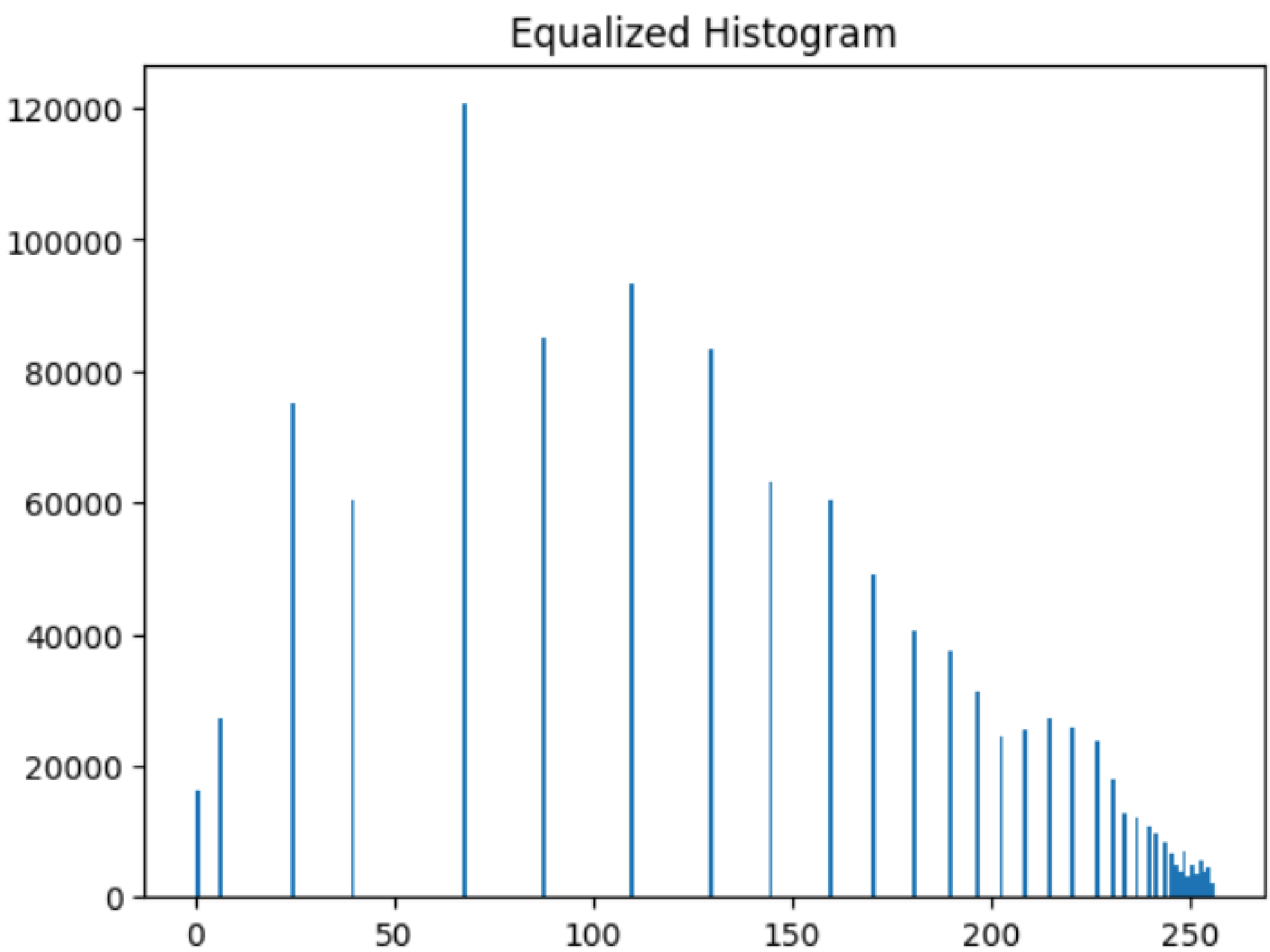
original_image



Histogram of Grayscale Image and any channel of Color Image

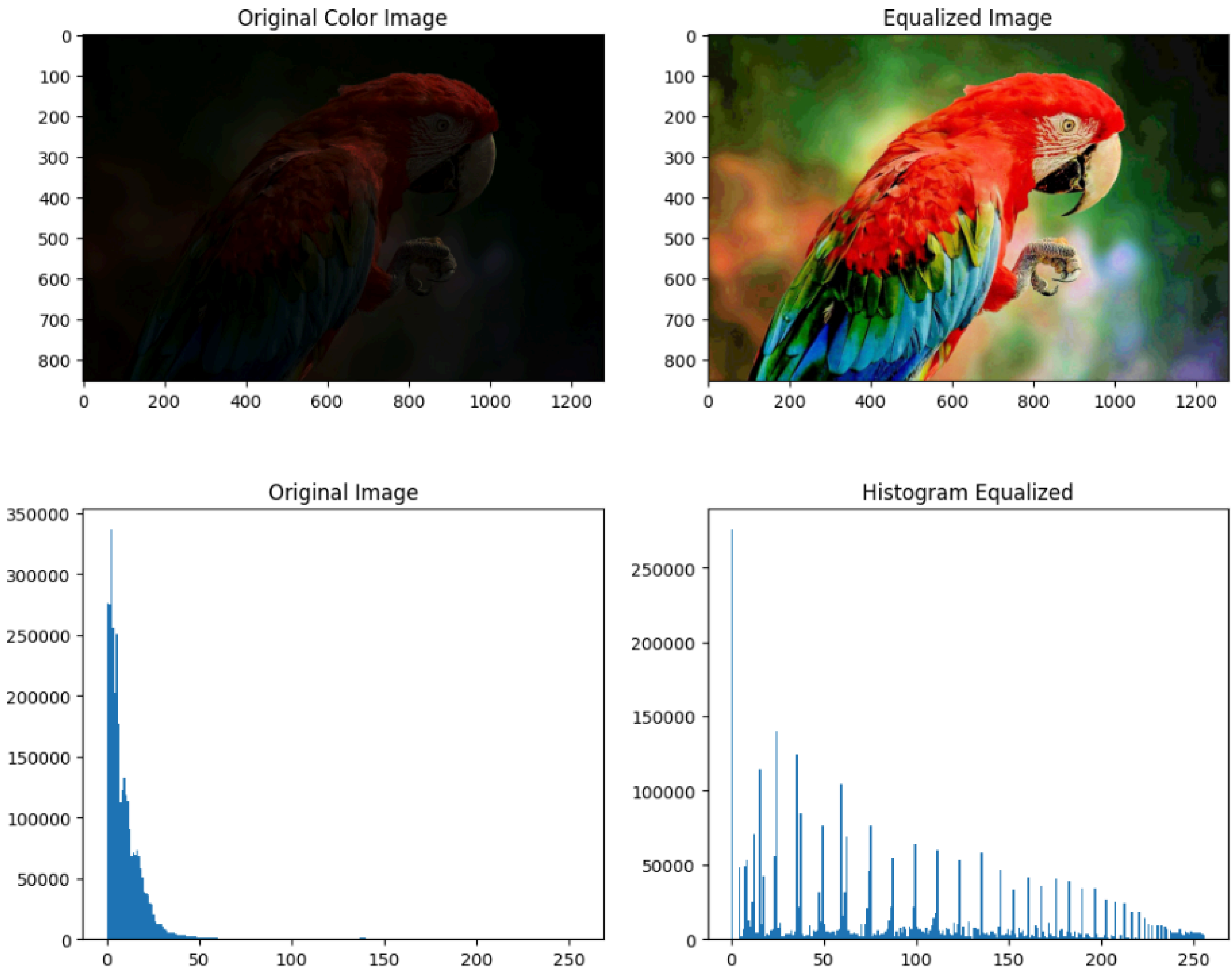



```
Text(0.5, 1.0, 'Equalized Histogram')
```



Histogram Equalization of Grayscale Image

```
Text(0.5, 1.0, 'Histogram Equalized')
```



Result:

Thus the histogram for finding the frequency of pixels in an image with pixel values ranging from 0 to 255 is obtained. Also,histogram equalization is done for the gray scale image using OpenCV.